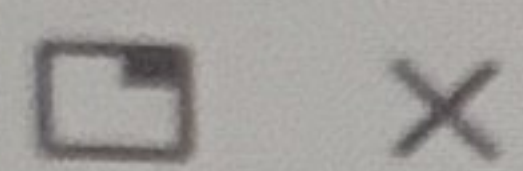


Files

[4]  
✓ 5s

# TRAIN &amp; EVALUATE

# =====

results = []

for name, model in models.items():

```
    pipe = Pipeline([
        ('prep', preprocessor),
        ('model', model)
    ])
```

pipe.fit(X\_train, y\_train)

y\_pred = pipe.predict(X\_test)

y\_prob = pipe.predict\_proba(X\_test)[:,-1]

```
    results.append({
        'Model': name,
        'Accuracy': accuracy_score(y_test, y_pred),
        'Precision': precision_score(y_test, y_pred),
        'Recall': recall_score(y_test, y_pred),
        'F1': f1_score(y_test, y_pred),
        'ROC_AUC': roc_auc_score(y_test, y_prob)
    })
```

print(f"\n{name}")

print(classification\_report(y\_test, y\_pred))

```
# =====
# RESULTS
# =====
```

results\_df = pd.DataFrame(results)

print("\nModel Comparison:")

```
print(results_df.sort_values(
    by='ROC_AUC',
    ascending=False
))
```

Random Forest

precision

recall

f1-score

support

Disk

65.52 GB available

Variables

Terminal



Search



DELL